Name

Mathematics Extension 1 Year 12

Properties of Functions, Relations and Graphs

Book 1

Version: 261021

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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-02 uses functions and relations to model, analyse and solve problems

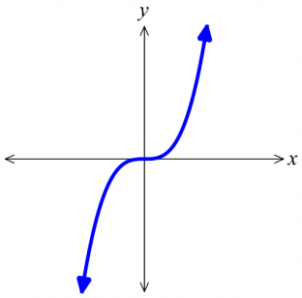
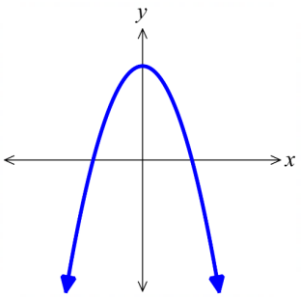
Properties of functions, relations and graphs

- Extend the definitions of domain and range to relations
- Recognise domains and ranges of functions and relations given in interval notation, as inequalities and as worded descriptions
- Determine and describe the domain and range of functions and relations, using interval notation, inequalities or worded descriptions
- Define a function to be even if its graph is unchanged under reflection in the y-axis, and odd if its graph is unchanged under rotation of $180°$ about the origin
- Develop and use the tests that a function $f(x)$ is odd if $f(-x) = -f(x)$ and a function $f(x)$ is even if $f(-x) = f(x)$
- Solve problems involving even and odd functions
- Use the composite function $f\big(g(x)\big)$, where the output of $g(x)$ becomes the input of $f(x)$
- Determine the equations of composite functions

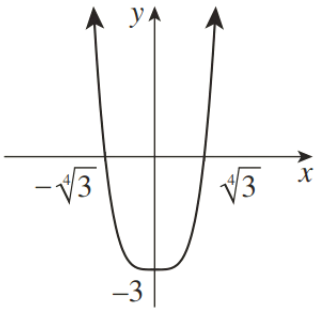
Odd and Even Functions

ⓘ Odd and Even Function

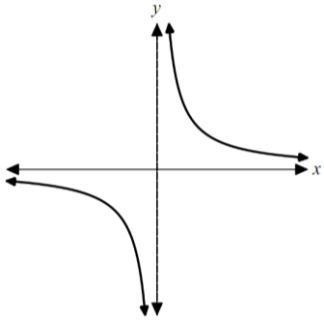
Functions can show two types of symmetry, helping us solve problems involving their graphs.

Odd Function	Even Function
180° rotational symmetry about the origin	Line symmetry across y-axis
	

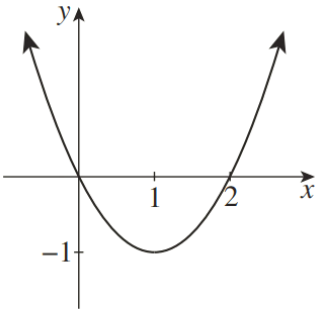
🔍 **Classify** graphs of functions as even, odd, neither



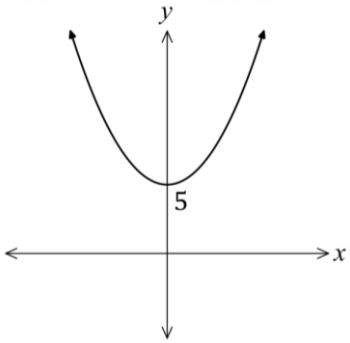
☐ Even
☐ Odd
☐ Neither



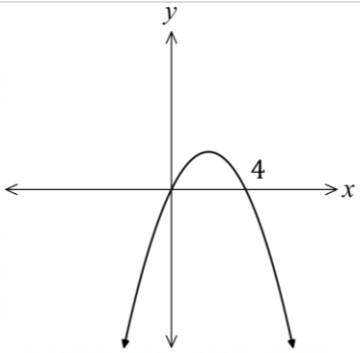
☐ Even
☐ Odd
☐ Neither



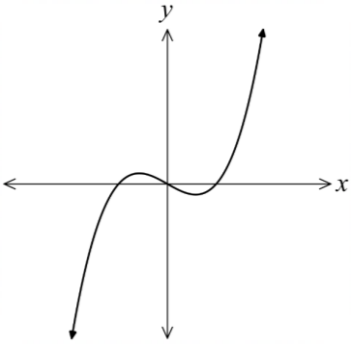
☐ Even
☐ Odd
☐ Neither



☐ Even
☐ Odd
☐ Neither



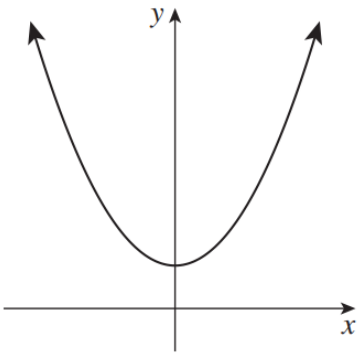
☐ Even
☐ Odd
☐ Neither



☐ Even
☐ Odd
☐ Neither

1 Classify these functions as even, odd, or neither.

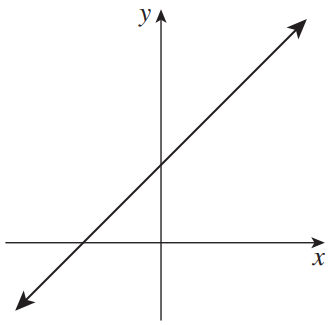
a



☐ Even
☐ Odd
☐ Neither

even

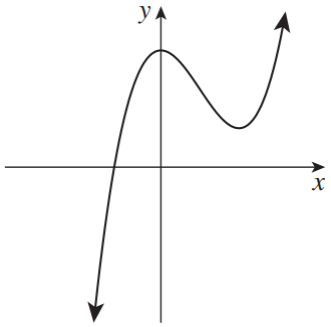
b



☐ Even
☐ Odd
☐ Neither

neither

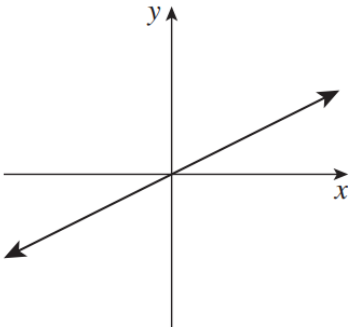
c



☐ Even
☐ Odd
☐ Neither

neither

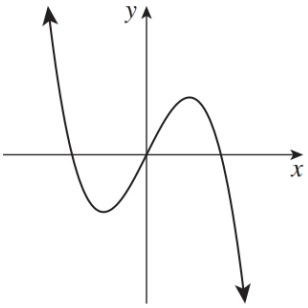
d



☐ Even
☐ Odd
☐ Neither

odd

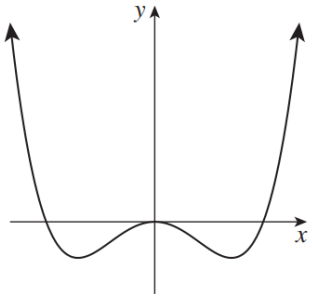
e



☐ Even
☐ Odd
☐ Neither

odd

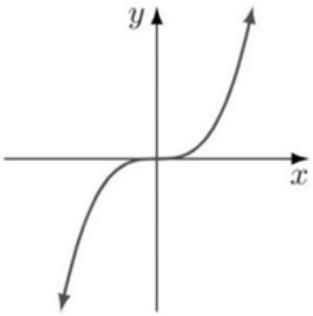
f



☐ Even
☐ Odd
☐ Neither

even

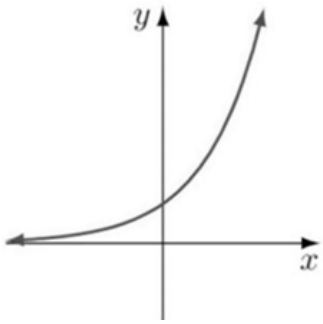
g



☐ Even
☐ Odd
☐ Neither

odd

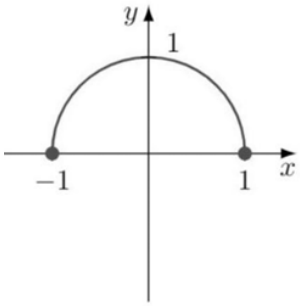
h



☐ Even
☐ Odd
☐ Neither

neither

i

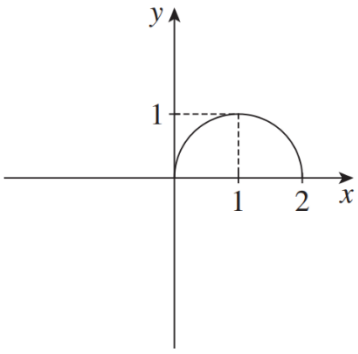


☐ Even
☐ Odd
☐ Neither

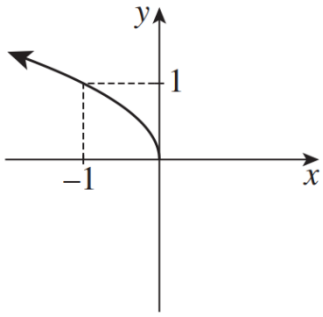
even

2 Sketch the second half of these graphs so they are even.

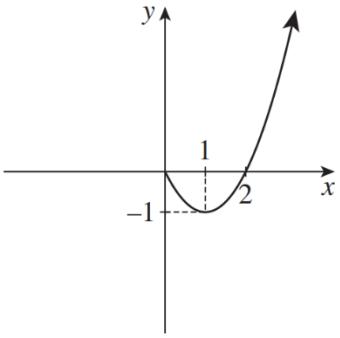
a



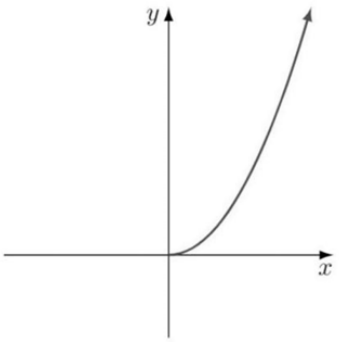
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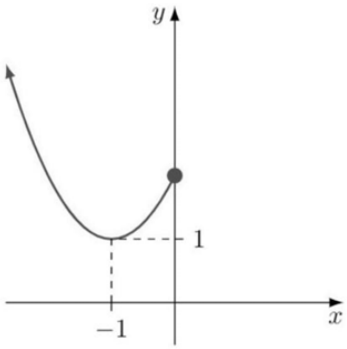
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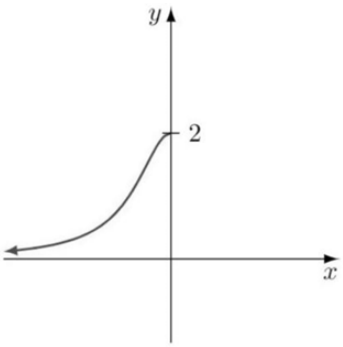
d



e

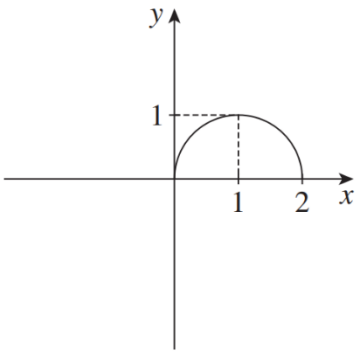


f

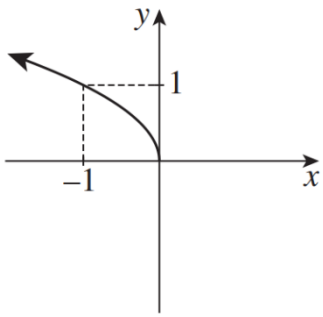


3 Sketch the second half of these graphs so they are odd.

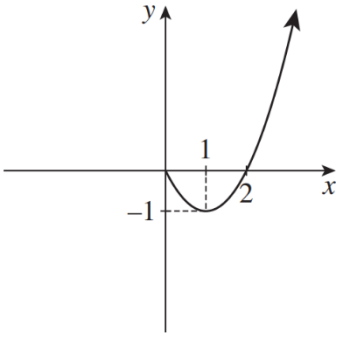
a



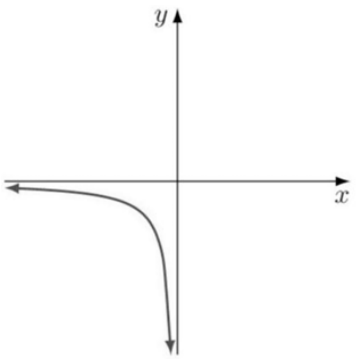
b



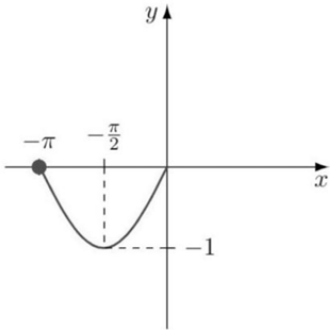
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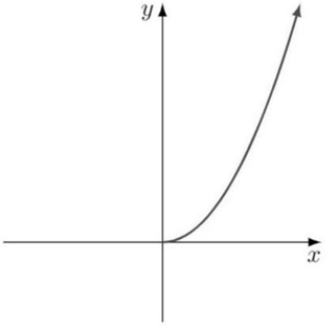
d



e



f



Testing Odd and Even Algebraically

Review

1 Find $f(-x)$ and $-f(x)$ for each of these functions.

a $f(x) = 3x - 2$

b $f(x) = x^2 - 4$

c $f(x) = x^2 - 2x$

d $f(x) = x^3 + x$

e $f(x) = 9 - x^2$

f $f(x) = x^2 + 5x + 6$

g $f(x) = \frac{x+3}{x}$

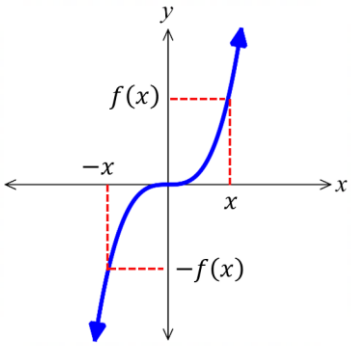
h $f(x) = \frac{x^3}{x^4 - x^2}$

i $f(x) = \frac{x-3}{x+3}$

ⓘ Testing Odd and Even Algebraically

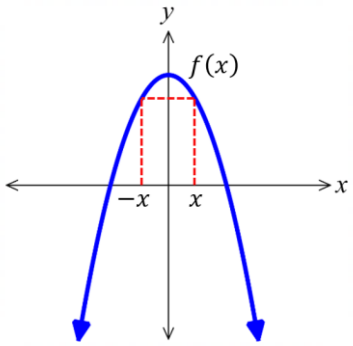
Odd Function

$f(-x) = -f(x)$



Even Function

$f(-x) = -f(x)$



1. Substitute in $f(-x)$.
2. Simplify as much as possible.
3. Even if $f(-x) = f(x)$.
   Odd if $f(-x) = -f(x)$.
   Neither if $f(-x)$ does not match $f(x)$ nor $-f(x)$.

Example Classify equations of functions as odd, even, or neither algebraically

$f(x) = x^4 - 3$

1. $f(-x) = (-x)^4 - 3$
2. $= x^4 - 3$
3. $= f(x)$

Even function.

$f(x) = x^3 - x$

1. $f(-x) = (-x)^3 - (-x)$
2. $= -x^3 + x$
3. $= -f(x)$

Odd function.

$f(x) = x^2 - 2x$

1. $f(-x) = (-x)^2 - 2(-x)$
2. $= x^2 + 2x$
3. Neither odd nor even.

Guided Practice

Determine whether these functions are odd, even, or neither.

a $f(x) = x^2 + 5$

even

b $f(x) = x(x - 4)$

neither

c $f(x) = x - x^3$

odd

d $f(x) = \frac{x^2+1}{x}$

odd

e $f(x) = \frac{x^3}{x^3-x}$

even

f $f(x) = \frac{x-3}{x+3}$

neither

1 Determine whether these functions are even, odd, or neither.

a $f(x) = 2x$

$f(-x) =$

odd

b $f(x) = x^2 + 1$

even

c $f(x) = x^2 + x$

neither

d $f(x) = x^3$

odd

e $f(x) = x - x^3$

odd

f $f(x) = \frac{1}{x}$

odd

g $f(x) = x^4 - 2x^2 + 1$

even

h $g(x) = x^3 - 3x$

odd

i $h(x) = x^3 + 3x^2 - 2$

neither

j $f(x) = x^2 - 9$

even

k $f(x) = x^3 - 25x$

odd

l $f(x) = x^3 + 5x^2$

neither

m $f(x) = 2^x$

neither

n $f(x) = \sqrt{3 - x^2}$

even

o $f(x) = \frac{1}{x^2+1}$

even

p $f(x) = \frac{4x}{x^2+4}$

odd

q $f(x) = x\sqrt{9 - x^2}$

odd

r $f(x) = \frac{x}{x^2-4}$

odd

2 Decide whether these functions are odd or even.
Check your answer by sketching the curve on DESMOS.

a $y = \frac{x^3}{x^4 - x^2}$

odd

b $y = \frac{1}{x^3 - 1}$

neither

c $y = \frac{3}{x^2 - 4}$

even

d $y = \frac{x - 3}{x + 3}$

neither

e $y = \frac{x^3}{x^5 - x^2}$

neither

f $y = (x - 2)^2$

neither

g $f(x) = 3^x - 3^{-x}$

odd

h $f(x) = 3^x + 3^{-x}$

even

i $f(x) = 3^x + x^3$

neither

3 For what positive integer values of n is the function $f(x) = kx^n$

a Even?

even values, $n = 2, 4, 6, 8, \ldots$

b Odd?

odd values, $n = 1, 3, 5, 7 \ldots$

4 **HSC Advanced Sample Question Band 4**

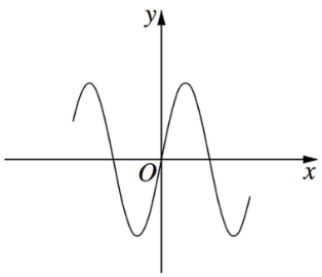
Determine whether $f(x) = 2x^3 - 5x$ is even, odd, or neither. Show all working.

odd

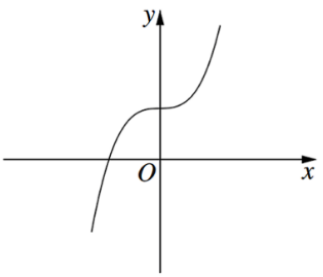
5 **2016 HSC Advanced Band 5**

Which diagram shows the graph of an odd function?

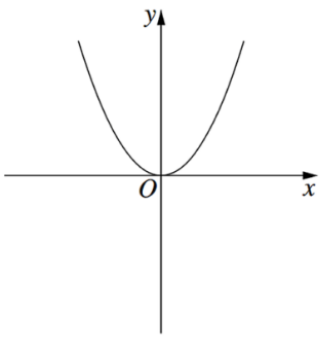
(A)



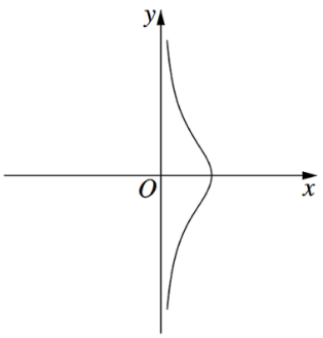
(B)



(C)



(D)



A

6 **2023 HSC Advanced Band 6**

Let $f(x)$ be any function with domain all real numbers.

Which of the following is an even function, regardless of the choice of $f(x)$?

(Hint: let $f(x)$ be a function that is neither, e.g. $f(x) = x + 2$)

A. $2f(x)$

B. $f(f(x))$

C. $(f(-x))^2$

D. $f(x)f(-x)$

D

7 Show that $f(x) = \dfrac{2^{2x}+1}{2^x}$ is even.

Composite Functions

ⓘ Composite Functions

A **composite** function is where the output of one function is inputted into another function.

We can consider any function that involves two or more operations as a composite function.

In $f(g(x))$, $g(x)$ is the inner function (evaluated first).

$f(x)$ is the outer function (evaluated last).

We can write composite functions with notation $f(g(x))$, $fg(x)$, $f \circ g$.

We read $f(g(x))$ as 'f of g'

🔍 **Investigation** Composite functions

Consider $f(x) = x^2$ and $g(x) = x + 5$

Find $g(3)$ ……

Insert the output of $g(3)$ back into $g(x)$.

We write this as ……, and the result is: ……

Now insert the output of $g(3)$ back into $f(x)$.

We write this as ……, and the result is: ……

Let's do the general case; what would be the result if we insert the output of $g(x)$ into $f(x)$?

$f(g(x)) =$

Check this works for $f(g(3))$

Example Determine the equation of a composite function

Given $f(x) = 2x + 5$ and $g(x) = 3 - 4x$, find the simplified expression for $f(g(x))$.

$$\begin{aligned} f(g(x)) &= f(3 - 4x) \\ &= 2(3 - 4x) + 5 \\ &= 6 - 8x + 5 \\ &= 11 - 8x \end{aligned}$$

Guided Practice

a Given $f(x) = 4x - 1$ and $g(x) = 2x + 7$, find $f(g(x))$.

$8x + 27$

b Given $f(x) = 5 - 2x$ and $g(x) = x^2 + 3$, determine $f(g(x))$ and $g(f(x))$.

$f(g(x)) = -2x^2 - 1$
$g(f(x)) = 4x^2 - 20x + 28$

c Given $f(x) = x^2 - 25$ and $g(x) = x + 4$
Solve $fg(x + 1) = gf(x)$

$x = -2.1$

Key Ideas

1. A composite function involves using the …… of the inner function as the …… to the outer function.

1 $f(x) = x + 2$.
Find the values of:

a $f(f(0))$

4

b $f(f(3))$

7

c $f(f(-1))$

3

d $f(f(-8))$

-4

e Find an expression for $f(f(x))$

$x + 4$

f Check your answers for **a – d**, how do they relate to your expression for $f(f(x))$?

The composite function is to add 4 to the value of x, which reflect the answers to part a – d.

g Find a value of x where $f(f(x)) = 0$.

$x = -4$

2 $f(x) = 2x$
Find the values of:

a $f(f(0))$

0

b $f(f(7))$

28

c $f(f(-3))$

-12

d $f(f(-11))$

-44

h Find an expression for $f(f(x))$

$4x$

i Check your answers for **a – d**, how do they relate to your expression for $f(f(x))$?

The composite function is to multiply the value of x by 4, which reflect the answers to parts a – d.

j Find a value of x where $f(f(x)) = 32$.

$x = 8$

3 $g(x) = 2 - x$. Show that $g(g(x)) = x$.

$g(x) = 2 - x$

$g(2 - x) = 2 - (2 - x)$

$\therefore g(2 - x) = x$

4 $h(x) = 3x - 5$. Find an expression for $h(h(x))$.

$9x - 20$

5 Let $f(x) = 3x + 4$ and $g(x) = 3 - 2x$.

a Find $f(g(2))$

1

b Find $g(f(2))$

-17

c Show that $f(g(x)) = -6x + 13$

$f(x) = 3x + 4$

$f(g(x)) = f(3 - 2x)$

$= 3(3 - 2x) + 4$

$= -6x + 13$

d Show that $g(f(x)) = -6x - 5$

$g(x) = 3 - 2x$

$g(f(x)) = g(3x + 4)$

$= 3 - 2(3x + 4)$

$= -6x - 5$

6 $f(x) = x + 1$ and $g(x) = 2x - 3$

Find the values of:

a $f(g(7))$

12

b $g(f(7))$

13

c $f(f(7))$

9

d $g(g(7))$

19

Find expressions for

e $f(g(x))$

$2x - 2$

f $g(f(x))$

$2x - 1$

g $f(f(x))$

$x + 2$

h $g(g(x))$

$4x - 9$

7 Given $f(x) = 2x + 7$ and $g(x) = 5x - 3$, find:

c $f(g(x))$

$10x + 1$

d $g(f(x))$

$10x + 32$

e $f(f(x))$

$4x + 21$

f $g(g(x))$

$25x - 18$

8 $f(x) = \sqrt{x}$ and $g(x) = x + 4$

a Find $f(g(x))$

$\sqrt{x+4}$

b Find $g(f(x))$

$\sqrt{x}+4$

c Find $g(g(x))$

$x + 8$

d What composite function would produce $\sqrt{x+8}$?

$f(g(g(x)))$

9 Given $p(x) = x^2 - 4$ and $q(x) = 2x + 3$, simplify:

a $p(q(x))$

$4x^2 + 12x + 5$

b $q(p(x))$

$2x^2 - 5$

c $p(p(x))$

$x^4 - 8x^2 + 12$

d $q(q(x))$

$4x + 9$

10 Given $h(x) = \frac{2}{x+3}$ and $m(x) = 4x - 1$, find:

g $h(m(x))$

$\frac{1}{2x+1}$

h $m(h(x))$

$\frac{5-x}{x+3}$

i $h(h(x))$

$\frac{2x+6}{3x+11}$

11 Find $f(g(x))$ and $g(f(x))$ for each of these function pairs.

a $f(x) = x - 2,\ g(x) = 2x + 5$

$f(g(x))$

$2x + 3$

$g(f(x))$

$2x + 1$

b $f(x) = 2x,\ g(x) = 3x$

$f(g(x))$

$6x$

$g(f(x))$

$6x$

c $f(x) = x^2,\ g(x) = x^3$

$f(g(x))$

x^6

$g(f(x))$

x^6

d $f(x) = \frac{1}{x},\ g(x) = \sqrt{x}$

$f(g(x))$

$\frac{1}{\sqrt{x}}$

$g(f(x))$

$\frac{1}{\sqrt{x}}$

12 $f(x) = \frac{x}{x+1}$, find $f(f(x))$.

$\frac{x}{2x+1}$

13 **HSC Sample Question Band 4**

Given $f(2) = 5, f(3) = 4, g(2) = 5, g(3) = 2, g(4) = 1.$

What is the value of $f(g(3))$?

$f(g(3)) = f(2) = 5$

14 HSC Sample Question Band 4

Which of these functions satisfies $f\big(f(x)\big) = x$?

A. $f(x) = 2 - x$

B. $f(x) = x^2$

C. $f(x) = 2\sqrt{x}$

D. $f(x) = x - 2$

A

15 HSC Sample Question Band 4

Which of these functions satisfies $f\big(f(x)\big) = x$?

A. $f(x) = x + 1$

B. $f(x) = x - 1$

C. $f(x) = \dfrac{x-1}{x+1}$

D. $f(x) = \dfrac{x+1}{x-1}$

D

16 Consider the graphs of $f(x)$ and $g(x)$ below.

a Evaluate $g(3)$.

b Hence evaluate $f\big(g(3)\big)$.

c Evaluate $g\big(f(0)\big)$.

d $f\big(f(1)\big)$.

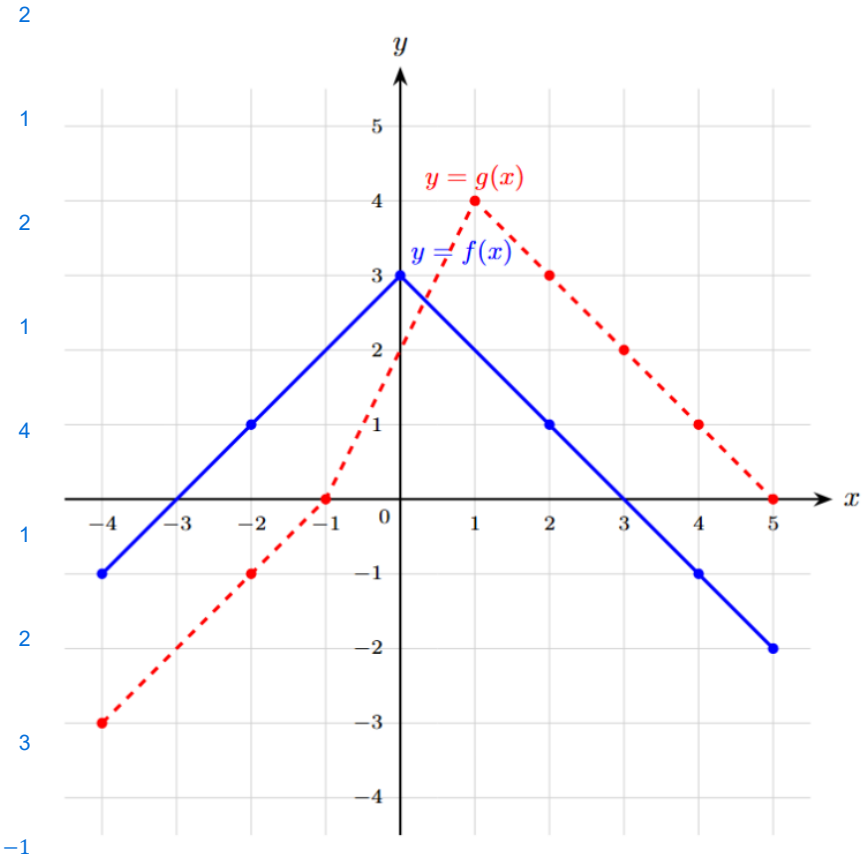
e $g\big(g(4)\big)$.

f $f\big(g(3)\big)$

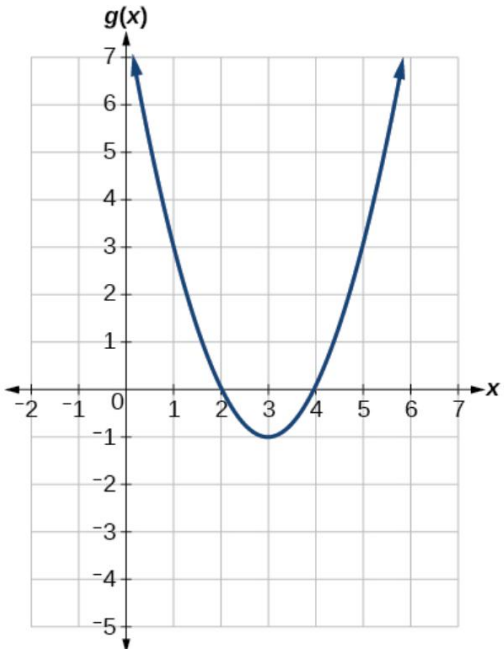
g $g\big(f(3)\big)$

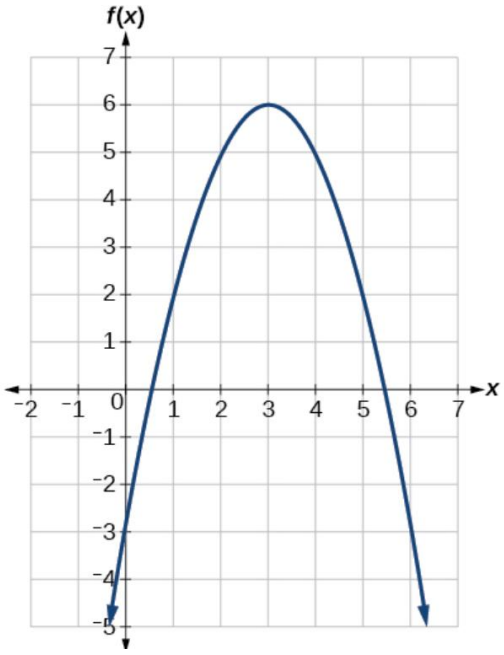
h $f\big(g(5)\big)$

i $g\big(f(5)\big)$



17 Using the graphs below, evaluate $f\big(g(1)\big)$ and $g\big(f(2)\big)$





$f\big(g(1)\big)$: $g(1) = 3, f(3) = 6$

$g\big(f(2)\big)$: $f(2) = 5, g(5) = 3$

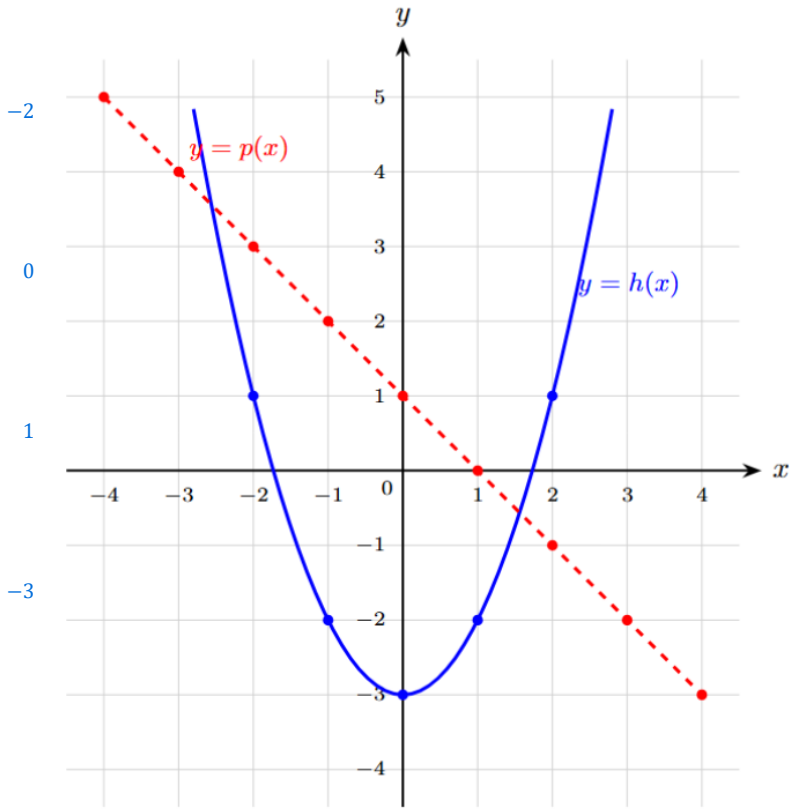
18 Consider the graphs of $p(x)$ and $h(x)$ below. Evaluate:

a $h\big(p(2)\big)$

b $p\big(h(2)\big)$

c $h\big(h(1)\big)$

d $p\big(p(-3)\big)$



19 $f(x) = x^2 - 3$ and $g(x) = x - 1$

a Find an expression for $\mathrm{gf}(x)$

b Find an expression for $\mathrm{fg}(x)$

c Solve the equation $\mathrm{gf}(x) = \mathrm{fg}(x)$

$$gf(x) = x^2 - 4$$
$$fg(x) = x^2 - 2x - 2$$
$$x^2 - 4 = x^2 - 2x - 2$$
$$2x = 2$$
$$x = 1$$

20 $g(x) = x - 12$ and $h(x) = x^2 - 10$

Solve the equation $\mathrm{gh}(x) = \mathrm{hg}(x)$

$$gh(x) = x^2 - 22$$
$$hg(x) = (x - 12)^2 - 10 = x^2 - 24x + 134$$
$$x^2 - 22 = x^2 - 24x + 134$$
$$24x = 156$$
$$x = 6.5$$

21 $f(x) = 2x + k$ and $g(x) = kx + 3$

$\mathrm{fg}(x) = 10x + m$

Find the values of k and m.

$$fg(x) = 2(kx + 3) + k = 2kx + 6 + k$$
$$2k = 10$$
$$k = 5$$
$$m = 6 + k = 11$$

22 $g(x) = 3x + 2$ and $h(x) = 6x + 5$

Show that $\mathrm{gh}(x) = \mathrm{hg}(x)$

$$gh(x) = 3(6x + 5) + 2 = 18x + 17$$
$$hg(x) = 6(3x + 2) + 5 = 18x + 17$$
$$\therefore gh(x) = hg(x)$$

23 $g(x) = \frac{2x+1}{x-1}$ and $h(x) = \frac{x+3}{x-5}$

Find an expression for $\mathrm{gh}(x)$

$$gh(x) = \frac{2 \cdot \frac{x+3}{x-5} + 1}{\frac{x+3}{x-5} - 1}$$
$$= \frac{2(x+3) + (x-5)}{(x+3) - (x-5)}$$
$$= \frac{3x+1}{8}$$

24 Let $f(x) = \frac{1}{4x-1}$ and $g(x) = 2^x$. Find the value of k such that $g\big(g(3)\big) = f(k)$.

$g\big(g(3)\big) = 2^{2^3} = 256$

$256 = \frac{1}{4k-1}$

$4k - 1 = \frac{1}{256}$

$4k = \frac{257}{256}$

$k = \frac{257}{1024}$

25 Let $f(x) = \sqrt{x}$ and $g(x) = -x^2 - 1$. Find $g\big(f(x)\big)$ and $f\big(g(x)\big)$.
Which composite function does not exist, and why?

$g\big(f(x)\big) = -x - 1$

$f\big(g(x)\big) = \sqrt{-x^2 - 1}$

Since $-x^2 - 1$ is always negative, $f\big(g(x)\big)$ is undefined.

26 Given that $f(x) = 4x - 3$ and $f(g(x)) = 12x + 5$, determine the algebraic formula for $g(x)$.

$f(g(x)) = 4(g(x)) - 3$
$4(g(x)) - 3 = 12x + 5$
$4(g(x)) = 12x + 8$
$g(x) = 3x + 2$

27 Two linear functions are defined as $f(x) = 2x + k$ and $g(x) = 3x - 4$, where k is a real constant.

a Find simplified algebraic expressions for $f(g(x))$ and $g(f(x))$.

$f(g(x)) = 2(3x - 4) + k = 6x - 8 + k$
$g(f(x)) = 3(2x + k) - 4 = 6x + 3k - 4$

b Determine the unique value of k for which $f(g(x)) = g(f(x))$ for all values of x.

For $f(g(x)) = g(f(x))$ to hold identically for all x:

$6x - 8 + k = 6x + 3k - 4$
$-8 + k = 3k - 4$
$-4 = 2k$
$k = -2$

28 Let $g(x)$ be an even function and $f(x)$ be any function. Let $h(x) = f\big(g(x)\big)$.

a Explain why $h(-x) = f\big(g(x)\big)$.

$h(-x) = f\big(g(-x)\big)$

since $g(x)$ is even, then $g(-x) = g(x)$

$\therefore h(-x) = f\big(g(x)\big)$

b Hence explain why the function $h(x)$ is even.

Since $h(-x) = f\big(g(x)\big) = h(x)$, it is even.

c Therefore, what can you say about the composition of two functions if the "inside" function is even?

It is even.

d Similarly, show that if $f(x)$ is even and $g(x)$ is odd, then $h(x) = f\big(g(x)\big)$ is even.

$h(-x) = f\big(g(-x)\big)$

since $g(x)$ is odd, then $g(-x) = -g(x)$

$h(-x) = f\big(-g(x)\big)$

since $f(x)$ is even, $f(-x) = f(x)$

$\therefore h(-x) = f\big(g(x)\big) = h(x)$

e Similarly, show that if $f(x)$ and $g(x)$ are both odd, then $h(x) = f\big(g(x)\big)$ is odd.

$h(-x) = f\big(g(-x)\big)$

since $g(x)$ is odd, then $g(-x) = -g(x)$

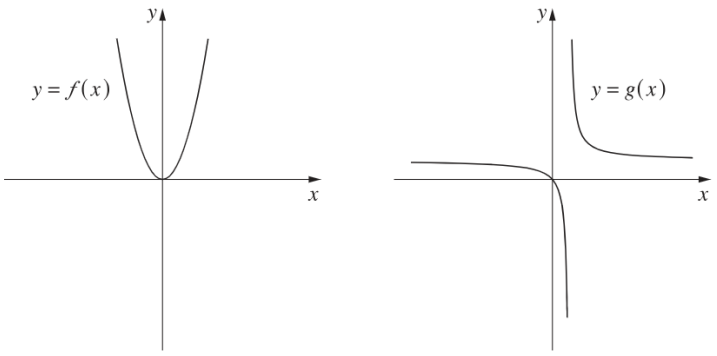
$h(-x) = f\big(-g(x)\big)$

since $f(x)$ is also odd, $f(-x) = -f(x)$

$\therefore h(-x) = -f\big(g(x)\big) = -h(x)$

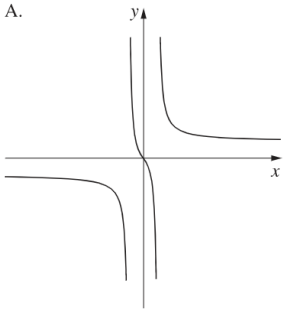
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The graphs of $y = f(x)$ and $y = g(x)$ are shown.

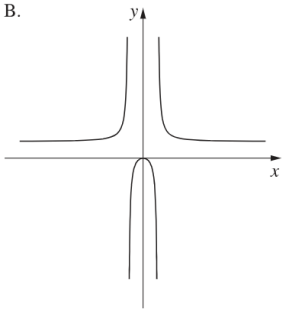


Which graph best represents $y = g\big(f(x)\big)$?

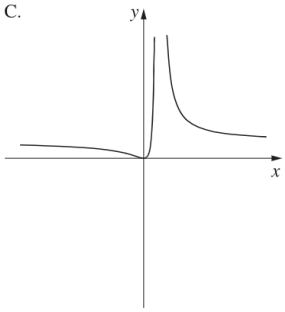
A.



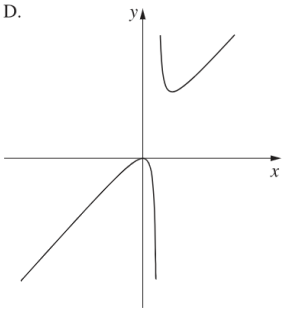
B.



C.



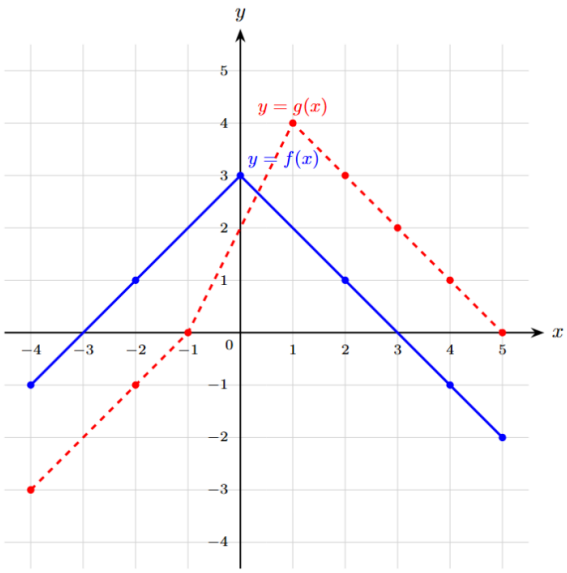
D.



Since $f(x)$ is even, $g(f(x))$ will also be even.

B

30 Consider once again the graph of $f(x)$ and $g(x)$ below.



a Find all values of x in the domain $[-4,5]$ that satisfy $f(g(x)) = 3$.

Consider $f(g(x)) = 3$

Look for points where graph f reaches height $y = 3$.

From the grid, $f\big(g(x)\big) = 3$ only when $g(x) = 0$.

Solve $g(x) = 0$:

$x = -1, 5$

b **[2-cycle fixed points]**

i Evaluate $ffff(x)$ for $x = 0$.

ii Evaluate $ffff(x)$ for $x = 3$.

iii Explain geometrically why the pair $\{0,3\}$ forms an alternating cycle under composition by f, and state other alternating cycles for f.

Repeated compositions of $x = 0$ and $x = 3$ keep alternating between 0 and 3.

An alternating cycle occurs when $f(a) = b$ and $f(b) = a$

Geometrically, this means (a, b) and (b, a) must lie on the graph of $y = f(x)$

These two points are mirror reflections across the line $y = x$

Therefore, all points with symmetry across $y = x$ on the graph of $f(x)$ will have this property, such as $\{1,2\}$ and $\{0.5,2.5\}$

By considering the graph, the solution set is all (x, y) pairs satisfying $y = 3 - x$ in the domain $x \in [0,3]$

Domain and Range of Composite Functions

ⓘ Domain and Range of Composite Functions

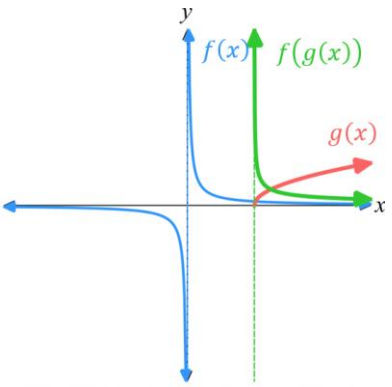
To find the **domain** of $f(g(x))$:

1. Write the equation of $f(g(x))$.
2. Determine the natural domain using zero denominators being undefined and square roots must be non-negative.

Example Determine domain and range of a composite function

$f(x) = \frac{1}{x}$

$g(x) = \sqrt{x-4}$



Find the domain of $f(g(x))$.

$f(g(x)) = \frac{1}{\sqrt{x-4}}$

$x - 4 > 0$

$x > 4$

Find the range of $f(g(x))$.

Since $\sqrt{x-4} > 0$, then $\frac{1}{\sqrt{x-4}} > 0$

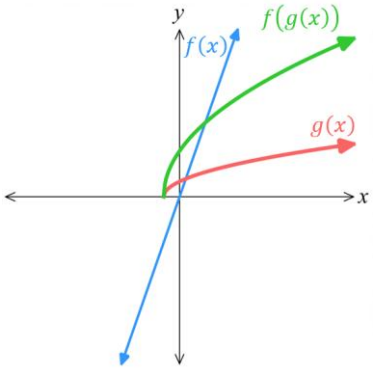
$\therefore f(g(x)) > 0$

Guided Practice

a

$f(x) = 3x$

$g(x) = \sqrt{x+1}$



Find $f(g(x))$.

$3\sqrt{x+1}$

Hence find the domain of $f(g(x))$.

$x \geq -1$

Hence find the range of $f(g(x))$.

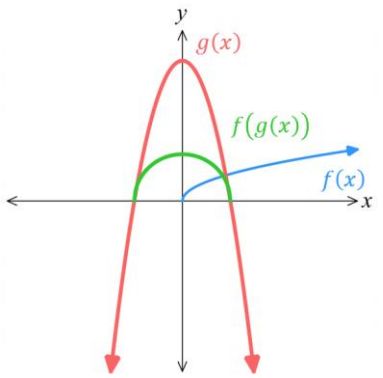
$f(g(x)) \geq 0$

b

$f(x) = \sqrt{x}$

$g(x) = 9 - x^2$

Find $f(g(x))$.



$\sqrt{9 - x^2}$

Hence find the domain of $f(g(x))$.

$-3 \leq x < 3$

Hence find the range of $f(g(x))$.

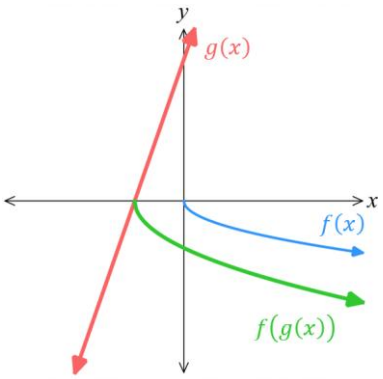
$0 \leq f(g(x)) \leq 3$

c

$f(x) = -\sqrt{x}$

$g(x) = 3x + 9$

Find $f(g(x))$.



$-\sqrt{3x + 9}$

Hence find the domain of $f(g(x))$.

$x \leq -3$

Hence find the range of $f(g(x))$.

$f(g(x)) \leq 0$

1 Write the composite function $f(g(x))$ and state its domain and range.

a $f(x) = x^2$ and $g(x) = x - 1$

$y = x^2 - 2x + 1$
D: $(-\infty, \infty)$
R: $[0, \infty)$

b $f(x) = x^3$ and $g(x) = x + 5$

$y = (x + 5)^3$
D: $(-\infty, \infty)$
R: $(-\infty, \infty)$

c $f(x) = \sqrt{x}$ and $g(x) = x - 2$

$y = \sqrt{x - 2}$
D: $[2, \infty)$
R: $[0, \infty)$

d $f(x) = -\sqrt{x}$ and $g(x) = 3x + 9$

$y = -\sqrt{3x + 9}$
D: $[-3, \infty)$
R: $(-\infty, 0]$

e $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$

$\sqrt{4 - x^2}$
D: $[-2,2]$
R: $[0,2]$

f $f(x) = -\sqrt{x}$ and $g(x) = 1 - x^2$

$-\sqrt{1 - x^2}$
D: $[-1,1]$
R: $[-1,0]$

2 $f(x) = \frac{2}{x-9}$ and $g(x) = x^2$.

a Find $f(g(x))$

$\frac{2}{x^2 - 9}$

b Find $g(f(x))$

$\frac{4}{(x-9)^2}$

c Compare the domain of $f(g(x))$ and $g(f(x))$.

$f(g(x))$ has domain $x \in \mathbb{R}, x \neq \pm 3$ while $g(f(x))$ has domain $x \in \mathbb{R}, x \neq 9$

3 $f(x) = x - 2$ and $g(x) = \sqrt{x}$

a Find $f(g(x))$.

$\sqrt{x} - 2$

b Hence state the domain of $f(g(x))$.

$x \geq 0$

c Find the domain of $g(f(x))$.

$x \geq 2$

4 $f(x) = 2x - 3,\ g(x) = \sqrt{x},\ h(x) = x^2 + 1$

Simplify these expressions, hence state the domain.

a $f(g(x))$

$2\sqrt{x} - 3, x \geq 0$

b $g(f(x))$

$\sqrt{2x - 3}, x \geq \frac{3}{2}$

c $h(f(x))$

$4x^2 - 12x + 10, x \in \mathbb{R}$

d $g(h(x))$

$\sqrt{x^2 + 1}, x \in \mathbb{R}$

5 If $f(x) = x^2 - 3$ and $g(x) = \sqrt{x - 2}$, find $f(g(x))$ and its domain.

$f(g(x)) = (\sqrt{x-2})^2 - 3 = x - 5$

Since the domain of $g(x)$ is restricted to $x \geq 2$, the domain of $f(g(x))$ is also restricted to $x \geq 2$

6 If $f(x) = \sqrt{x}$ and $g(x) = 25 - x^2$, find $f\big(g(x)\big)$ and its domain.

$f\big(g(x)\big) = \sqrt{25 - x^2}$

D: $-5 \leq x \leq 5$

R: $0 \leq y \leq 5$

7 Given $f(x) = \sqrt{x^2 - 9}$ and $g(x) = x + 5$

a Find integers c and d such that $f\big(g(x)\big) = \sqrt{(x + c)(x + d)}$

$f\big(g(x)\big) = \sqrt{(x + 5)^2 - 9}$

$= \sqrt{x^2 + 10x + 16}$

$= \sqrt{(x + 2)(x + 8)}$

$c = 2, d = 8$ (or other way around)

b Hence state the domain for which $f\big(g(x)\big)$ is defined.

$(x + 2)(x + 8) \geq 0$

$\therefore x \leq -8, x \geq 2$